

Sampa Halder

Houston, Texas • +1-806-224-4105 •

sampahldr@gmail.com • shalder@cougarnet.uh.edu • www.linkedin.com/in/sampa-halder

RESEARCH SUMMARY

Researcher with expertise in bio-fabrication, 3D bioprinting, biomaterials, and advanced manufacturing, committed to advancing translational biomedical research and healthcare innovation through interdisciplinary collaboration.

EDUCATION

University of Houston , Houston, Texas, USA	<i>Aug 2027 (Expected)</i>
PhD in Industrial Engineering (in progress), CGPA: 4.00/4.00	
Research Area: Bio-fabrication, 3D bioprinting, Electrospinning	
Texas Tech University , Lubbock, Texas, USA	<i>Aug 2024</i>
Master of Science in Industrial Engineering, CGPA: 4.00/4.00	
Bangladesh University of Engineering and Technology , Dhaka, Bangladesh	<i>Apr 2019</i>
Bachelor of Science in Industrial Engineering, GPA: 3.53/4.00	

RESEARCH EXPERIENCE

Graduate Research Assistant , University of Houston, Texas, USA	<i>Aug 2024 – Present</i>
- Investigating the printability of bioinks and mechanical properties of hydrogel scaffolds for biomedical applications using volumetric printing, extrusion-based printing. - Developing biomimetic hydrogel scaffolds embedded with microtubes for enhanced perfusion and tissue engineering. - Optimizing bio-fabrication parameters using artificial neural networks (ANN) and multi-response optimization (MRO). - Conducting research on nano- and micro-scale material development using electrospinning and electrospraying techniques. - Working with 3D metal printing using the Laser Engineering Net Shaping (LENS) process to study alloy properties for biomedical implementation. - Collaborating on interdisciplinary projects integrating engineering and medical sciences for biomedical innovation.	

PROFESSIONAL EXPERIENCE

Teaching Assistant , Texas Tech University	<i>Sep 2022 – July 2024</i>
- Graded exams and assignments, supervised student projects, and provided academic support - Delivered lectures on Biomedical Innovation and Advanced Manufacturing	
Lecturer , International University of Business, Agriculture and Technology, Bangladesh	<i>Oct 2021 – July 2022</i>
- Designed course materials, delivered lectures, and conducted interactive classes - Supervised undergraduate theses, mentoring students in research execution	
Senior Supply Chain Officer , Rahimafrooz Renewable Energy Limited, Bangladesh	<i>Oct 2020 – Sep 2021</i>
- Managed procurement of local and international materials for production and projects - Strategically planned supply chains to ensure seamless product availability and delivery	
Operations Coordinator , Square Fashions Limited, Bangladesh	<i>Sep 2019 – Sep 2020</i>
Analyzed processes and compared performance metrics to identify improvement opportunities	

SKILLS

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| <ul style="list-style-type: none">Soft Skills: Project Management, Problem Solving, Team Collaboration, Leadership, Written/Verbal and Interpersonal Communications, AdaptabilityLanguage: English and Bengali (Native) | <ul style="list-style-type: none">Programming Languages: MATLAB, Python, C and RSoftware: AutoCAD, Inventor, MS Excel, MATLAB, Simio, Arena, CATIA, Solidworks, Ansys, Minitab |
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PROFESSIONAL SERVICE

- Session Chair**, “Bioprinting, Biomedical Devices & Wearable Sensors,” IISE Annual Conference & Expo 2026

- **Reviewer**, Materials Today Communications, Elsevier (2025)
- **Reviewer**, International Journal of Mechanical Engineering Education (2025)

HONORS AND AWARDS

- Best M&D student paper at IISE conference, Canada (2024)
- “Best track paper award” at IEOM conference, Bangkok (2019)
- “Best track paper award” at IEOM conference, Pilsen (2019)
- 100% scholarship in PTAK Supply Chain Competition (2017)
- Gold, Silver, and Bronze award winner in Duke of Edinburgh’s Award (2018, 2017, 2016)

CERTIFICATION

- Certified Internal Auditor Training in ISO 9001: 2015 (*Management Consulting Solution*, 2021)
- Certified Internal Auditor Training in ISO 45001:2018 (*iota Consulting BD*, December 2021)
- Supervised Machine Learning: Regression and Classification (*Coursera*, May 2024)
- Data Analytics for Lean Six Sigma (*Coursera*, July 2020)
- Introduction to Data Analytics for Business (*Coursera*, May 2020)

PUBLICATIONS

1. Zhang, N., Qavi, I., Araneda, M., Rumi, S. S., Abidi, N., **Halder, S.**, & Tan, G. Z. (2025). **3D bioprinting of self-strengthening living materials using cellulose nanofiber-producing bacteria in sodium alginate hydrogel.** *Bioprinting*, e00443. <https://doi.org/10.1016/j.bprint.2025.e00443>
2. **Halder, S.**, & Tan, G. Z. (2025). **Musculoskeletal tissue regeneration using fibrous biomaterials.** In *Tissue Repair and Regeneration: Elucidating Cellular and Molecular Mechanisms*. Springer. https://doi.org/10.1007/978-3-031-93677-7_14
3. **Halder, S.**, LeFebvre, L., Mann, M., & Tan, G. Z. (2025). **Collaborative engineering-medical class for non-specialists: Fostering interdisciplinary engaged learning for biomedical innovation.** *International Journal of Mechanical Engineering Education*. <https://doi.org/10.1177/03064190251388930>
4. Qavi, I., **Halder, S.**, & Tan, G. (2024). **Optimization of printability of bioinks with multi-response optimization (MRO) and artificial neural networks (ANN).** *Progress in Additive Manufacturing*. <https://doi.org/10.1007/s40964-024-00828-1>.
5. **Halder, S.**, Qavi, I., & Tan, G. (2024, June). **Investigating the effect of aligned microtube density on the mechanical strength of hydrogel scaffolds.** In *Proceedings of the ASME 2024 19th International Manufacturing Science and Engineering Conference* (Vol. 1: Additive Manufacturing; Advanced Materials Manufacturing; Biomanufacturing; Life Cycle Engineering, V001T03A011). Knoxville, Tennessee, USA: ASME. <https://doi.org/10.1115/MSEC2024-125393>.
6. Qavi, I., **Halder, S.**, Du, D., & Tan, G. (2024, May). **Machine learning enhanced high fidelity and high viability bioprinting using rheological and compositional predictors.** Paper presented at the *IISE Annual Conference & Expo 2024*. https://doi.org/10.21872/2024IISE_7125.
7. Mann, M., Qavi, I., **Halder, S.**, & Tan, G. (2024). **Developing Problem Solving Competency for Future Engineers in Medicine.** *Journal of Higher Education Theory and Practice*, 24(6). <https://doi.org/10.33423/jhetp.v24i6.7017>
8. Zhang, N., Qavi, I., **Halder, S.**, & Tan, G. (2023). **Biomimetic hydrogel scaffolds embedded with porous microtubes as perfusion channels.** *Manufacturing Letters*, 35(Supplement), 184–193. <https://doi.org/10.1016/j.mfglet.2023.08.009>
9. Tan, G., **Halder, S.**, & LeFebvre, L. (2023, June). **Cultivate the problem exploration skills for biomedical innovation.** Paper presented at the 2023 *ASEE Annual Conference & Exposition*, Baltimore, Maryland. <https://doi.org/10.18260/1-2--42836>.
10. Sarwar, F., Islam, F., Sakib, M., & **Halder, S.** (2019, July). **Optimizing a Solid Waste Management Model using Particle Swarm Optimization.** In *3rd European International Conference on Industrial Engineering and Operations Management*, <https://doi.org/10.46254/EU03.20190427>.

11. Sarwar, F., Islam, F., Sakib, M., & Halder, S. (2019, July). **Identifying Barriers of Lean Six Sigma Implementation in RMG Sector: A Case Study.** In *3rd European International Conference on Industrial Engineering and Operations Management*, <https://doi.org/10.46254/EU03.20190390>.
12. Sarwar, F., Islam, F., Sakib, M., & Halder, S. (2019, March). **Identifying Drivers of Lean Six Sigma Implementation in the Process Industries: A Case Study.** In *9th Annual International Conference on Industrial Engineering and Operations Management*, <https://doi.org/10.46254/AN09.20190376>

CONFERENCE PRESENTATION

1. Halder, S., Karim, M. R., Hai, T. B., & Tan, G. (2026, May). **Controlling structural stability through bioresin optical properties for tomographic volumetric printing.** Oral presentation at the IISE Annual Conference & Expo 2026.
2. Halder, S., Qavi, I., & Tan, G. (2024, June). **Investigating the effect of aligned microtube density on the mechanical strength of hydrogel scaffolds.** Oral presentation at the ASME 2024 International Manufacturing Science and Engineering Conference, Knoxville, Tennessee, USA.